

DIF1908 Método de Integración por partes.

$$\int x \cos(x^2) dx \rightarrow \text{Substitution Simple. v/s } \int x \cos(x) dx$$

el método: $(uv)' = u'v + uv'$ (derivada del producto)

↓
Integrando.

$$\int (uv)' dx = \int u'v dx + \int uv' dx$$

$$uv' = \int v u' dx + \int u v' dx$$

$$uv = \int v du + \int u dv$$

formula de integración
por
partes

$$\int u dv = uv - \int v du$$

$$\int v du = uv - \int u dv$$

$$\int x \cos(x) dx$$

$$= x \sin(x) - \int \sin(x)$$

~~$$\int x \cos(x) dx = x \sin(x) - \int \sin(x)$$~~

1) $u = x$
 $dv = \cos(x) dx$; $v = \sin(x)$

3) $\int x \cos(x) dx = x \sin(x) - \int \sin(x) dx$
 $= x \sin(x) + \cos(x) + C$

L I A T E

i) $\log(x)$

ii) Inversa Trigonometric $\left[\begin{matrix} \arcsin(x) \\ \arccos(x) \\ \arctan(x) \end{matrix} \right]$

iii) Algebraic Terms $\left[\begin{matrix} \text{polynomials} \end{matrix} \right]$

iv) Trigonometric $(\sin, \cos, \tan)$

v) Exponentes.

Example a) $\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C$
 $= e^x (x - 1) + C$
 $u = x$
 $dv = e^x dx \rightarrow v = e^x$

Example b) $\int x \ln(x) dx$ $u = \ln(x)$ $dv = x dx \rightarrow \frac{x^2}{2} = v$
 $du = \frac{1}{x} dx$

$$= \ln(x) \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx$$

$$= \frac{x^2}{2} \ln(x) - \frac{1}{2} \int x dx$$

$$= \frac{\ln(u) x^2}{2} - \frac{x^2}{4} + C$$

Ex 1: $\int x^2 e^x dx$ $u = x^2 \rightarrow du = 2x dx$
 $dv = e^x dx \rightarrow v = e^x$

$$= x^2 e^x - \int e^x 2x dx$$

$$= x^2 e^x - 2 \int x e^x dx = x^2 e^x - 2(x-1)e^x + C$$

$$\int x^3 e^x dx = x^3 e^x - \int x^2 e^x dx$$

$$= x^3 e^x - (x^2 e^x - \int e^x 2x dx)$$

$$= x^3 e^x - (x^2 e^x - 2e^x(x-1))$$

$$= x^3 e^x - x^2 e^x + 2e^x(x-1) + C$$

$u = x^3 \rightarrow du = 3x^2 dx$
 $dv = e^x dx \rightarrow v = e^x$

$\int (x^2 + 3x) e^{2x} dx$ $u = x^2 + 3x \rightarrow du = 2x + 3 dx$
 $dv = e^{2x} dx \rightarrow \frac{e^{2x}}{2} = v$

$$= \frac{(x^2 + 3x) e^{2x}}{2} - \frac{1}{2} \int e^{2x} (2x + 3) dx \quad \text{CASA}$$

$u = 2x + 3 \rightarrow du = 2 dx$
 $dv = e^{2x} dx \rightarrow v = \frac{e^{2x}}{2}$

$$= \frac{(x^2 + 3x) e^{2x}}{2} - \frac{1}{2} \left[\left(\frac{e^{2x}}{2} (2x + 3) \right) - \int \frac{e^{2x}}{2} dx \right]$$

$$= \frac{(x^2 + 3x) e^{2x}}{2} - \left(\frac{e^{2x}(2x+3)}{4} \right) + \frac{1}{2} \int e^{2x} dx = \frac{(x^2 + 3x) e^{2x}}{2} - \frac{e^{2x}(2x+3)}{4} + \frac{e^{2x}}{4} + C$$

$$\int x^2 \ln(x) dx = \frac{x^3}{3} \ln(x) - \frac{1}{3} \int x^2 dx = \frac{x^3}{3} \ln(x) - \frac{x^3}{9} + C$$

$$u = \ln(x) \rightarrow du = \frac{1}{x} dx$$

$$dv = x^2 dx \rightarrow v = \frac{x^3}{3}$$

Ejemplo 7. $\int x^3 \ln(x) dx = \frac{x^4}{4} \ln(x) - \int \frac{x^4}{4} \frac{1}{x} dx$

$$u = \ln(x) \rightarrow du = \frac{1}{x} dx$$

$$dv = x^3 dx \rightarrow v = \frac{x^4}{4}$$

$$= \frac{x^4}{4} \ln(x) - \frac{1}{4} \int \frac{x^4}{x} dx = \frac{x^4}{4} \ln(x) - \frac{1}{4} \int x^3 dx$$

$$= \frac{x^4}{4} \ln(x) - \frac{x^4}{16} + C$$

Ejemplo 8.

$$\int \ln(x) dx = x \ln(x) - \int x \cdot \frac{1}{x} dx$$

$$u = \ln(x) \rightarrow du = \frac{1}{x} dx = x \ln(x) - \int dx$$

$$dv = dx \rightarrow v = x = x \ln(x) - x = x(\ln(x) - 1) + C$$

9. $\int x \arctan(x) dx$ $u = \arctan(x) \rightarrow du = \frac{1}{x^2+1} dx$

$$dv = x dx \Rightarrow v = \frac{x^2}{2}$$

$$= \arctan(x) \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot (x^2+1)^{-1} dx$$

$$= \frac{x^2}{2} \arctan(x) - \frac{1}{2} \int \frac{x^2}{x^2+1} dx$$

PENDIENTE

Ex 10) $\int \arcsin(x) dx = x \arcsin(x) - \int \frac{x}{\sqrt{1-x^2}} dx = x \arcsin(x) - \int \frac{du}{2\sqrt{u}} = x \arcsin(x) - \sqrt{1-x^2} + C$

$u = \arcsin(x) \rightarrow du = \frac{1}{\sqrt{1-x^2}} dx$
 $dv = dx \rightarrow v = x$

$\int x(\sqrt{1-x^2})^{-1/2} dx$
 $u = 1-x^2$
 $du = -2x dx$

$(a^x)' = a^x \ln(a)$
 $= x \arcsin(x) - \sqrt{1-x^2} + C$

Ex 11) $\int x \sec^2(x) dx$
 $u = x \rightarrow du = dx$
 $dv = \sec^2(x) dx \rightarrow v = \tan(x)$

$= x \tan(x) - \int \tan(x) dx$
 $= x \tan(x) + \ln|\cos(x)| + C$

12) $\int x^2 \cos(x) dx = x^2 \sin(x) - \int \sin(x) 2x dx$
 $u = x^2 \rightarrow du = 2x dx$
 $dv = \cos(x) dx \rightarrow v = \sin(x)$

$= x^2 \sin(x) - 2 \int \sin(x) x dx$
 $= x^2 \sin(x) - 2 \left(-x \cos(x) + \int \cos(x) dx \right) = x^2 \sin(x) + 2x \cos(x) - 2 \sin(x) + C$
 $= x^2 \sin(x) - 2 \left(x \cos(x) \right) + \int \cos(x) \frac{x^2}{2} dx$
 $= x^2 \sin(x) + 2x \cos(x) - 2 \sin(x) + C$

Existe intégrales, qui au trait de intégr par parties des veces, nous a permis le intégral original $\rightarrow$ "intégrales cicliques".

B) $\int e^x \cos(x) dx = e^x \cos(x) + \int e^x \sin(x) dx = e^x \cos(x) + e^x \sin(x) - \int e^x \cos(x) dx$

$u = \cos(x) \rightarrow du = -\sin(x)$
 $dv = e^x dx \rightarrow v = e^x$

$2 \int e^x \cos(x) dx = e^x \cos(x) + e^x \sin(x)$
 $\int e^x \cos(x) dx = \frac{e^x (\cos(x) + \sin(x))}{2} + C$